When Does Market Access Become Local Development? Platform Integration, Comparative-Advantage Realization, and Local Value Capture

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Abstract

Platform integration can lower local consumer prices while shifting trade away from locally embedded channels. It also gives local producers access to a larger market, but that opportunity need not become local production and income. This paper asks when these two sides of platform integration jointly produce local development. Its theoretical object is the access–capture problem developed in a companion short paper. New Structural Economics supplies the structural backbone on the local-development side: endowments shape latent comparative advantage; hard and soft infrastructure determine actual comparative advantage and firm viability; and organizational embeddedness determines whether production and complementary-service payments remain local. The model therefore separates consumer access, producer access, local channel displacement, comparative-advantage realization, and local capture formation. Its target results are distinct thresholds for nominal local income and consumption-equivalent real income, with the latter strictly lower because consumers benefit from the price decline even when first-round local receipts fall. Public facilitation addresses an explicit shared-infrastructure or coordination failure in an LCA-consistent activity. This internal draft fixes the paper's theoretical architecture; proposition statements are not claims of proof until the corresponding derivations are completed in the appendix.

1 Introduction

A common market does not connect every region to the national economy in the same way. Platform integration can make goods produced elsewhere cheaper and more accessible to local consumers. At the same time, it can displace local wholesale, retail, logistics, and transaction services. For local producers, the same platform can enlarge the potential customer base. Whether that producer-side opportunity becomes local output, employment, service income, and profits depends on the region's production structure and the transaction conditions under which its firms operate.

The paper asks:

> Why does platform-enabled market integration broadly improve consumer access, yet translate into local production and income only when a region's latent comparative advantage can be realized through appropriate local organization and infrastructure?

This formulation fixes the division of labor between the paper's two theoretical components. The access–capture problem is the object to be explained. New Structural Economics (NSE) provides the local structural block: it disciplines industry choice, the conversion from latent to actual comparative advantage, firm viability, organizational embeddedness, local linkages, and the facilitating-state boundary. The platform block supplies the market-access and channel-substitution shocks, while the companion paper supplies the income-propagation block. The paper is therefore neither a general theory of structural transformation with a platform added at the end nor a version of the companion model in which "local capability" is simply renamed comparative advantage.

The intended contribution has three parts. First, consumer-side and producer-side market access are modeled separately, even though both improve with platform integration. Second, local value capture is derived from the location and ownership of production and transaction-service payments, rather than imposed as a regional parameter; production and local complementary services are jointly generated by the NSE block. Third, the paper connects the NSE distinction among latent comparative advantage, actual comparative advantage, viability, and realization to the companion paper's distinction between nominal local income and consumption-equivalent real income.

2 The Access–Capture Benchmark

2.1 Consumer access and channel substitution

Consider one focal region embedded in a larger national market. A representative local household consumes a channel-composite good supplied through a locally embedded channel A and an external platform channel P:

$$C_M = \left[a_A^{1/\eta} C_A^{(\eta-1)/\eta} a_P^{1/\eta} C_P^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)}, \qquad \eta > 1.$$

The associated price index and platform expenditure share are

$$P_M(z) = \left[a_A p_A^{1-\eta} a_P p_P(z)^{1-\eta}\right]^{1/(1-\eta)}$$

and

$$s_P(z) = \frac{a_P p_P(z)^{1-\eta}}{a_A p_A^{1-\eta} + a_P p_P(z)^{1-\eta}}.$$

Platform integration is indexed by z and satisfies $p_P'(z) < 0$. Hence

$$s_P'(z) > 0, \qquad P_M'(z) < 0.$$

This block establishes a consumer-access gain and a shift in transaction organization. It does not, by itself, determine whether local firms can sell to the national market.

2.2 Payment location

Let ℓ_A and ℓ_P denote the locally received value generated by one unit of expenditure through the two channels. They are accounting outcomes, not independent behavioral parameters. Each is constructed from mutually exclusive payments to commodity producers, wholesale and retail services, logistics and fulfilment services, platform services, and locally owned profits. In the benchmark case,

$$\ell_A > \ell_P,$$

because a larger share of the traditional channel's transaction services is locally provided, whereas platform fees and part of the platform's organizational services accrue outside the focal region.

For a fixed first-round expenditure base X_i, the local transaction-service component is

$$V_i^{\text{channel}}(z) = X_i \left[\big(1 - s_P(z)\big)\ell_A + s_P(z)\ell_P\right].$$

Thus channel substitution can lower local receipts even while it lowers the consumer price index. This is the benchmark capture loss that the production-side model must offset; it is not the paper's final result.

3 Endowments and Latent Comparative Advantage

3.1 Production costs

Region i has endowment vector $E_i = (K_i, L_i)$. A competitive background sector 0 anchors local opportunity costs (w_i, r_i). Candidate industry j has factor intensity θ_j and unit production cost

$$c_{ij}^P = c_j(w_i, r_i; \theta_j).$$

The superscript P identifies production cost before industry-specific transaction costs. This distinction is essential: NSE does not define comparative advantage as the inverse of a single industry's absolute cost.

3.2 Strict relative-cost definition

Let h be an external reference region and let 0 be the internal benchmark industry. Define the production-cost comparison

$$\mathcal{C}_{ij}^P = \frac{c_{ij}^P / c_{i0}^P}{c_{hj}^P / c_{h0}^P}.$$

Region i has latent comparative advantage (LCA) in industry j when

$$\mathcal{C}_{ij}^P < 1.$$

This is a double-relative production-cost statement. It says that, before industry-specific transaction costs and infrastructure constraints, industry j is relatively better suited to region i than to the reference region. It does not imply that firms in i can profitably produce and sell the good under the existing business environment.

For clarity, the paper maintains four distinct concepts:

Concept	Meaning in the model
LCA	Relative production-cost advantage implied by endowments and sectoral factor intensities, before industry-specific transaction constraints.
ACA	Relative total-cost advantage after the prevailing hard and soft infrastructure, transaction costs, and organizational environment are taken into account.
Viability	A firm's ability to cover actual production, transaction, and fixed costs without permanent protection.
Realized comparative advantage	The equilibrium manifestation of comparative advantage through viable entry, production, and sales. It is an outcome, not an alternative name for LCA.

4 The NSE Structural Block

4.1 Producer access, comparative advantage, and viability

Let $M(z)$ measure the national demand accessible to a local producer, with $M'(z) > 0$. Channel r entails transaction cost $t_{ijr}(G_i)$ and fixed entry or organizational cost $f_{ijr}(G_i)$, where G_i is a parsimonious measure of the relevant shared infrastructure. The total unit cost is

$$c_{ijr}^{T} = c_{ij}^{P} + t_{ijr}(G_i),$$

and operating profit can be written as

$$\pi_{ijr}(z, G_i) = M(z) b_j\big(c_{ijr}^{T}\big) - f_{ijr}(G_i), \qquad b_j'(\cdot) < 0.$$

The firm enters the national market if

$$\max_{r} \pi_{ijr}(z, G_i) \geq 0.$$

This condition converts the potential represented by LCAinto an actual producer response. It also shows why LCA, ACA, viability, and realized comparative advantage cannot be collapsed into one index. LCAis a relative production-cost statement; ACAincorporates the prevailing transaction environment; viability requires the firm to cover its actual costs without persistent protection; and entry is the equilibrium realization of these conditions.

4.2 Organizational embeddedness and local linkages

On the producer side, an entrant chooses between external platform organization P and locally embedded organization L. The locally embedded consumer channel A in the previous section is not a third producer-side organization. The choice is

$$r_{ij}^{*} \in \arg\max_{r \in \{P, L\}} \{0, \pi_{ijr}(z, G_i)\}.$$

Mode P uses marketing, settlement, fulfilment, or supply-chain services provided outside the focal region. Mode L obtains the corresponding services from local providers. For each mode, the same service bill is decomposed as

$$H_{ijr} = H_{ijr}^{\text{local}} + H_{ijr}^{\text{external}}.$$

The first component enters local service income and the second is an external payment. Consequently, organizational embeddedness is not an independent retention parameter: it is the payment-incidence consequence of the firm's profit-maximizing organization.

In this baseline, local linkages mean only that complementary services required by the candidate industry and its platform sales are supplied locally. They do not introduce endogenous supplier varieties, agglomeration externalities, or cumulative causation.

4.3 Why infrastructure is in the model

G_i is introduced only for infrastructure or organizational services with a specific economic role: they reduce a transaction or fixed cost that a potentially viable industry faces when reaching the integrated market. In the policy section, public provision is permitted only if this input is shared, indivisible, coordination-intensive, or otherwise not fully internalized by a single entrant. Generic "capacity" that directly raises local retention is not admissible.

Target result 1a: Comparative-advantage realization

Under monotonicity and single-crossing conditions, there is an infrastructure threshold

$$G_E^* = G_E^*(\mathcal{C}_{ij}^P, M)$$

above which a local firm enters. Stronger LCAshould reduce the infrastructure required for entry. The result must be derived from the profit condition, rather than imposed through an increasing capability function.

Target result 1b: Organizational embeddedness

If local organization has a higher fixed cost but lowers the relevant variable service cost or external payment, the profit difference

$$\Delta\pi_{ij}^{L-P} = \pi_{ijL} - \pi_{ijP}$$

can generate an organizational-embedding threshold G_L^* or M_L^*. This result enters the baseline only if crossing the threshold changes H_{ijr}^{local}, $H_{ijr}^{\text{external}}$, and therefore the nominal or real local-income threshold.

5 Local Value Capture and Income

5.1 First-round local income

For one common set of transactions, first-round local income is

$$V_i = W_i^{\text{local}} + \Pi_i^{\text{local}} + S_i^{\text{local}} + T_i^{\text{local}} - C_i^{\text{policy}}.$$

The components are local labor income, locally owned net operating surplus, local transaction-service value added, net local fiscal receipts when policy is present, and the real resource cost of policy. Each payment appears once. The corresponding local retention share is

$$\omega_i = \frac{V_i}{\text{transaction value generated by the same transaction set}}.$$

The change induced by platform integration is organized as

$$\begin{aligned}\Delta V_i = \Delta V_i^{\text{producer}} &+ \Delta V_i^{\text{local services}} \\ &- \Delta V_i^{\text{agent displacement}} - \Delta V_i^{\text{external payments}}.\end{aligned}$$

The four terms respectively capture local entry, output, and outward sales; locally supplied platform complements; the loss of locally embedded channel services; and payments to the platform and outside organizational services. The NSE structural block generates the first term through viability and entry, and the second and fourth terms through organizational choice and service payment incidence. Only local-agent displacement is generated entirely by the consumer-side channel-substitution block. The substantive NSE extension therefore concerns both production and the local organizational structure through which transaction value is captured.

5.2 The companion-paper income block

The access–capture companion paper provides a conditional local-income propagation block. Denote its exact mapping by

$$Y_i = \mathcal{M}_i(\omega_i)V_i.$$

The final version will substitute the companion paper's exact fixed-point formula after matching the transaction denominator and avoiding double counting of producer and service payments. Consumption-equivalent local income is

$$R_i = \frac{Y_i}{P_M(z)^{\alpha}},$$

so that

$$\frac{d\ln R_i}{dz} = \frac{d\ln Y_i}{dz} - \alpha\frac{d\ln P_M(z)}{dz}.$$

The propagation block amplifies the first-round local-income consequence. It does not replace the production-side entry problem, and it must not be added as a second demand feedback inside the firm's resource constraint.

6 Target Equilibrium Results

Target result 2: Nominal local-income threshold

The model must derive a unique G_Y^* satisfying

$$\frac{dY_i}{dz} \geq 0 \quad \Longleftrightarrow \quad G_i \geq G_Y^*.$$

Economically, infrastructure and LCAjointly determine whether producer and local-linkage gains offset local-agent displacement and external platform payments. NSE affects both the positive production margin and the spatial incidence of complementary-service payments.

Target result 3: Real-income threshold

The consumer price decline implies a distinct threshold G_R^* satisfying

$$\frac{dR_i}{dz} \geq 0 \quad \Longleftrightarrow \quad G_i \geq G_R^*.$$

When both thresholds are interior, the target ordering is

$$G_R^* < G_Y^*.$$

This ordering generates three economically distinct regions:

$$\begin{cases} G_i < G_R^* : & \text{nominal and consumption-equivalent local income both fall,} \\ G_R^* \leq G_i < G_Y^* : & \text{nominal local income falls but the price gain raises real income,} \\ G_i \geq G_Y^* : & \text{production and local-linkage gains make both measures rise.} \end{cases}$$

Stronger LCAshould lower both thresholds because a given improvement in producer access and infrastructure induces a larger viable supply response and can support locally embedded complementary services. This statement requires an increasing-differences or equivalent single-crossing condition; it cannot be inferred merely by relabeling local capability.

7 The Facilitating State

The facilitating state is conceptually central to the NSE block but analytically downstream of the positive equilibrium results. Public action is justified only when potentially viable firms in an LCA-consistent industry cannot individually supply a shared input or coordinate complementary investments. Facilitation then lowers a real fixed or transaction cost:

$$f'_{ijr}(G_i) < 0 \quad \text{or} \quad t'_{ijr}(G_i) < 0,$$

at a convex resource cost to the local economy.

This intervention differs from protection. Facilitation improves the region's ability to realize a production-cost advantage and, where relevant, to form locally embedded complementary services. Protection raises the relative cost of the outside channel or shields firms that may not be viable. The model does not assume that every infrastructure project is beneficial or that facilitation always dominates every temporary support instrument. The policy result is conditional on three elements: an independently defined LCA, an explicit market failure preventing comparative-advantage realization or organizational embeddedness, and benefits that exceed the policy's real resource cost.

8 Scope and Extensions

The baseline deliberately excludes endogenous migration, agglomeration externalities, endogenous national market size, dynamic organizational capital, platform price setting, and a full multi-region general equilibrium. These mechanisms are not required to answer the access–capture question. New Economic Geography becomes relevant only in a later paper if endogenous firm location and market size produce cumulative causation, multiplicity, or path dependence.

The baseline retains only the minimal P–L comparison required to endogenize local-service and external-payment incidence. An earlier model contract studied three full producer-side thresholds: entry through an external platform, stand-alone profitability of a local organizational system, and switching from the platform to that system. The complete three-threshold structure may be restored as an extension only if it changes G_Y^* or G_R^* without introducing an additional state variable.

9 Conclusion

The paper's fixed subject is not the general formation of comparative advantage. It is the local development incidence of platform-enabled market integration. NSE provides the local structural backbone: endowments determine latent comparative advantage; the actual infrastructure and transaction environment determine ACAand viability; and organizational embeddedness determines whether complementary-service payments remain local. Combined with the consumer price effect, local-channel displacement, and the companion-paper income block, this mechanism can explain why equalized access coexists with unequal local capture.

A Proof and Accounting Appendix

This appendix records the exact completion standard for the internal theory draft. A target result in the main text becomes a formal proposition only after the corresponding item below contains assumptions, a complete derivation, and boundary cases.

A.1 Consumer-access block

1. Derive the channel demands, $s'_P(z) > 0$, and $P'_M(z) < 0$ from the CES expenditure problem.
2. State the admissible parameter region and the limit cases $s_P \to 0$ and $s_P \to 1$.

A.2 Comparative-advantage definitions

1. Derive local factor opportunity costs from the background sector and regional endowments.
2. Show how changes in the relevant endowment ratio affect $\mathcal{C}_{ij}^P$ for industries with different factor intensities.
3. Keep LCA, ACA, viability, and realized comparative advantage distinct throughout the derivation.

A.3 Entry, organization, and local-income thresholds

1. Solve the producer's entry problem and derive G_E^*.
2. Solve the P–L producer-organization choice and derive the conditions for G_L^* or M_L^*.
3. Keep consumer channel A distinct from producer organization L.
4. Construct one complete payment-flow table for channels A and P, local producer sales, and the local/external components of complementary services.
5. Verify that organizational choice changes both local service income and external payments; otherwise remove it from the baseline.
6. Derive the monotonicity and boundary conditions required for a unique G_Y^*.
7. Derive G_R^* and prove $G_R^* < G_Y^*$ when both thresholds are interior.
8. Report parameter regions in which either threshold does not exist or lies outside the feasible support.

A.4 Companion-paper mapping

The final appendix must reproduce the exact income-propagation equation used in the companion paper and provide a one-to-one map between:

1. its autonomous-income term and the producer-side payments derived here;
2. its local-retention denominator and the transaction set used here;
3. its consumer price index and $P_M(z)$;
4. its nominal and consumption-equivalent income definitions and (Y_i, R_i) here.

This check is required to rule out double counting of local production, transaction services, and induced household spending.

A.5 Policy and robustness

The policy proof must state the market failure, government budget, real resource cost, and incidence of any platform or local-service payment. It must also identify whether the intervention relaxes a constraint on comparative-advantage realization, organizational embeddedness, or both. Robustness work is limited to partial localization of platform-service income and multiple candidate industries unless an additional mechanism changes one of the core threshold results.